Civil Engineering Department
National Institute of Technology Kurukshetra

Name of Examination: B. Tech. (Civil) III Semester IInd Mid-sem Exam
Time: **50 min** Max. Marks: **45**

Note: Attempt any Five questions including Question nos. 1 and 2. All questions carry 09 marks each. You are expected to spend not more than 10 minutes on any question.

1. Derive Euler's equation of motion and hence deduce the Bernoulli's equation. State clearly the assumptions made.

2. (A) Explain **any three of the** following:
   i. An orificemeter and a venturimeter
   ii. Velocity of Approach
   iii. A sharp crested and a broad crested weir
   iv. Prandtl Pitot tube

   (3)/4 min

   (B) Show that a Cippollitte weir is a special kind of 'rectangular notch with no end contraction' effect

   (6)/6 min

3. In a smooth pipe of uniform diameter 25 cm, a pressure of 50 kPa was observed at section 1 which was at elevation 10.00m. At another section 2 at elevation 12.00m the pressure was 20 kPa and the velocity was 1.25 m/s. Determine the direction of flow and the head loss between these two sections. The fluid in the pipe is water.

4. An oil of relative density 0.9 flows through a vertical pipe of diameter of 10 cm. The flow is measured by a 20 cm x 10 cm venturimeter. The throat is 10 cm above the inlet section. A differential U–tube manometer containing mercury is connected to the throat and the inlet. If $C_d = 0.99$, what is the flow for a manometer reading of 9 cm?

5. Oil flows through a 25 mm diameter orifice under a head of 5.5 m at a rate of 3 L/s. The jet strikes a wall 1.5 m away and 120 mm vertically below the centreline of the contracted jet. Calculate the coefficient of velocity, contraction and discharge.

6. A discharge of 100 L/s is to be measured by a triangular notch of vertex angle 60°. What would be the head over the vertex of the notch? If the accuracy of reading the head is 1 mm what error in discharge can be expected at this level? Assume $C_d = 0.58$.

THE END